

Problem 5.13

Show that $I_b = I_e + \frac{I_e}{72}$ and $I_e = I_b - \frac{I_b}{73}$. Thus, the Banker's Rule is more favorable to the investor than the actual/actual rule.

Using exact simple interest,

$$I_e = Pr \left(\frac{t}{365} \right).$$

Using Banker's Rule,

$$I_b = Pr \left(\frac{t}{360} \right).$$

Now compare the two expressions:

$$\frac{I_b}{I_e} = \frac{Pr \left(\frac{t}{360} \right)}{Pr \left(\frac{t}{365} \right)} = \frac{365}{360}.$$

Since

$$\frac{365}{360} = 1 + \frac{5}{360} = 1 + \frac{1}{72},$$

we get

$$I_b = I_e \left(1 + \frac{1}{72} \right) = I_e + \frac{I_e}{72}.$$

Next, solve for I_e :

$$I_b = I_e \left(\frac{73}{72} \right).$$

Thus,

$$I_e = I_b \left(\frac{72}{73} \right).$$

Since

$$\frac{72}{73} = 1 - \frac{1}{73},$$

it follows that

$$I_e = I_b \left(1 - \frac{1}{73} \right) = I_b - \frac{I_b}{73}.$$

Therefore,

$$I_b > I_e,$$

so the Banker's Rule is more favorable to the investor than the actual/actual rule.